

# Ngoc Minh Quang Nguyen

✉ [quang@stat.cmu.edu](mailto:quang@stat.cmu.edu) 🔗 [qntkhvn](#) 🌐 [qntkhvn.netlify.app](#)

## EDUCATION

**Carnegie Mellon University**

PhD, Statistics & Data Science

2022 – present

**Loyola University Chicago**

MS, Applied Statistics

2020 – 2021

**Wittenberg University**

BS, Mathematics & Data Science

2017 – 2020

## PUBLICATIONS

### JOURNAL ARTICLES

R. Yurko, **Q. Nguyen**, K. Pelechris. NFL Ghosts: A framework for evaluating defender positioning with conditional density estimation. 2026. *Annals of Applied Statistics* **20**(1), 873–892. [doi:10.1214/25-AOAS2132](#); [arXiv:2406.17220](#).

**Q. Nguyen**, R. Yurko. A multilevel model with heterogeneous variances for snap timing in the National Football League. 2025. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*. [doi:10.1093/jrssa/qnaf185](#); [arXiv:2502.16313](#).

**Q. Nguyen**, R. Jiang, M. Ellingwood, R. Yurko. Fractional Tackles: Leveraging Player Tracking Data for Within-Play Tackling Evaluation in American Football. *Scientific Reports (Nature Portfolio)* **15**, 2148. 2025. [doi:10.1038/s41598-025-85993-1](#); [arXiv:2403.14769](#).

**Q. Nguyen**, R. Yurko, G. J. Matthews. Here Comes the STRAIN: Analyzing Defensive Pass Rush in American Football with Player Tracking Data. *The American Statistician* **78**(2), 199–208. 2024. [doi:10.1080/00031305.2023.2242442](#); [arXiv:2305.10262](#).

### Editors' Choice Collection.

**Q. Nguyen**, G. J. Matthews. Filling the Gaps: A Multiple Imputation Approach to Estimating Aging Curves in Baseball. *Journal of Sports Analytics* **10**(1), 77–85. 2024. [doi:10.3233/JSA-240744](#); [arXiv: 2210.02383](#).

N. M. Lemon, L. K. Taylor, M. A. Rech, **Q. Nguyen**, G. J. Matthews, G. Lew, S. Lovett. Utility of D-dimer in predicting pulmonary embolism in patients with COVID-19 presenting to the emergency department. *Journal of the American College of Emergency Physicians Open* **5**(4), e13237. 2024. [doi:10.1002/emp2.13237](#).

B. S. Baumer, G. J. Matthews, **Q. Nguyen**. Big Ideas in Sports Analytics and Statistical Tools for their Investigation. *Wiley Interdisciplinary Reviews: Computational Statistics* **15**(6), e1612. 2023. [doi:10.1002/wics.1612](#); [arXiv:2301.04001](#).

**Q. Nguyen**. Poisson Modeling and Predicting English Premier League Goal Scoring. *New England Journal of Statistics in Data Science* **1**(1), 95–101. 2023. [doi:10.51387/21-NEJSDS1](#); [arXiv:2105.09881](#).

**Q. Nguyen**, H. Butler, G. J. Matthews. An Examination of Olympic Sport Climbing Competition Format and Scoring System. *Journal of Data Science* **20**(2), 156–167. 2022. [doi:10.6339/22-JDS1042](#); [arXiv:2111.05310](#).

### PREPRINTS & WORKING PAPERS

**Q. Nguyen**, R. Yurko. Bayesian multilevel step-and-turn models for evaluating player movement in American football. 2026+. [arXiv:2603.17866](#).

**Q. Nguyen**, R. Yurko. A Bayesian circular mixed-effects model for explaining variability in directional movement in American football. 2025+. [arXiv:2507.06122](#).

**Q. Nguyen**, P. Campana, N. M. D. Niezink. Co-evolution of cooperation and conflict among organized crime groups: Evidence from the United Kingdom. 2025+.

## OTHER

**Q. Nguyen**, R. Yurko (collaboration between CMSAC and NFL). Introducing the NFL Draft Tier Chart. NFL Football Operations. 2026. <https://operations.nfl.com/updates/football-ops/introducing-the-nfl-draft-tier-chart>

**Q. Nguyen**, R. Yurko (collaboration between CMSAC and NFL). Building an NFL Draft Value Chart. NFL Football Operations. 2026. <https://operations.nfl.com/updates/football-ops/building-an-nfl-draft-value-chart>

**Q. Nguyen**, R. Yurko. Down, set, hut! Explaining variability in snap timing on plays with motion. NFL Big Data Bowl 2025. <https://www.kaggle.com/code/tindata/down-set-hut>

**Q. Nguyen**, L. Jiang, M. Ellingwood, R. Yurko. Momentum-based fractional tackles. NFL Big Data Bowl 2024. <https://www.kaggle.com/code/tindata/momentum-based-fractional-tackles>

G. J. Matthews, **Q. Nguyen**. STRAIN: Sacks, Tackles, Rushing, Aggression INDEX. NFL Big Data Bowl 2023. <https://www.kaggle.com/code/statsinthewild/strain-sacks-tackles-rushing-aggression-index>

B. S. Baumer, **Q. Nguyen**, G. J. Matthews. CRAN Task View: Sports Analytics. The Comprehensive R Archive Network. 2022. <https://cran.r-project.org/web/views/SportsAnalytics.html>

## ACADEMIC EXPERIENCE

### INSTRUCTOR

#### Carnegie Mellon University

36-315: Statistics Graphics and Visualization

Summer 2026

Summer Undergraduate Research Experience in Statistics

Summer 2024, Summer 2025

#### Loyola University Chicago

STAT 103: Fundamentals of Statistics (3 sections)

Spring 2022

### GRADUATE TEACHING ASSISTANT

#### Carnegie Mellon University

36-460/660: Sports Analytics

Spring 2024, Spring 2025, Spring 2026

36-468/668: Text Analysis

Fall 2023, Fall 2024, Fall 2025

Summer Undergraduate Research Experience in Statistics

Summer 2023

36-402/608: Advanced Methods for Data Analysis

Spring 2023

36-401/607: Modern Regression

Fall 2022

## OTHER

#### Wittenberg University

Math Workshop Head Tutor

2018 – 2020

Faculty Aide

2018 – 2020

#### Clarkston Senior High School

S.U.M. (Student Understanding Math) Assistant

2017

## RESEARCH EXPERIENCE

#### Carnegie Mellon University

2023 – present

Graduate Researcher (Advisor: Ron Yurko)

#### Carnegie Mellon University

2022 – 2023

Advanced Data Analysis Project (Advisor: Nynke Niezink)

#### Loyola University Chicago

2020 – 2021

Graduate Research Assistant (Advisor: Gregory J. Matthews)

#### Wittenberg University

2019 – 2020

Senior Honors Thesis Research (Advisor: Douglas M. Andrews)

## PRESENTATIONS

### TALKS

**Q. Nguyen**, R. Yurko. A Bayesian circular mixed-effects model for evaluating change of direction in American football. *Joint Statistical Meetings*. Boston, MA. August 2026. (Contributed)

**Q. Nguyen**, R. Yurko. NFL CoD: A Bayesian circular mixed-effects model for explaining variability in directional movement in American football. *New England Symposium on Statistics in Sports*. Cambridge, MA. September 2025. (Invited)

**Q. Nguyen**. Advances in pass rush and change of direction evaluation in American football with player tracking data. *U.S. Olympic and Paralympic Performance Innovation Summit*. Colorado Springs, CO. August 2025. (Invited)

**Q. Nguyen**. NFL CoD. *Uncontrolled Variables: A Science & Comedy Show*. Chicago, IL. August 2025. (Invited)

**Q. Nguyen**, R. Yurko. Down, set, hut! Explaining variability in snap timing on plays with motion. *Joint Statistical Meetings*. Nashville, TN. August 2025. (Contributed)

**Q. Nguyen**, R. Yurko. Step-and-turn modeling of individual player movement in American football. *Conference on Predictive Inference in Sports*. Ames, IA. June 2025. (Invited)

**Q. Nguyen**, R. Yurko. Explaining variability in player turning behavior in the National Football League. *CMSAC Football Analytics Workshop*. Pittsburgh, PA. May 2025. (Invited)

**Q. Nguyen**. Advances in pass rush and tackling evaluation in American football with player tracking data. *2024 Joint Statistical Meetings*. Portland, OR. August 2024. (Invited)

**Q. Nguyen**, L. Jiang, M. Ellingwood, R. Yurko. Fractional tackles. *UConn Sports Analytics Symposium*. Storrs, CT. April 2024. (Invited)

**Q. Nguyen**, L. Jiang, M. Ellingwood, R. Yurko. Momentum-based fractional tackles. *NFL Big Data Bowl 2024*. Indianapolis, IN. February 2024. (Invited)

**Finalist (Top 5 + \$12,500 prize + Presented at 2024 NFL Scouting Combine)**

**Q. Nguyen**. Opportunities in sports analytics with player tracking data: A case study of NFL pass rush evaluation. *Data Science Seminar, Loyola University Chicago*. Chicago, IL. November 2023. (Invited)

**Q. Nguyen**, R. Yurko, G. J. Matthews. Here Comes the STRAIN: Analyzing Defensive Pass Rush in American Football with Player Tracking Data. *New England Symposium on Statistics in Sports*. Cambridge, MA. September 2023. (Invited, featured talk)

G. J. Matthews, **Q. Nguyen**. STRAIN: Sacks, Tackles, Rushing, Aggression INdex. *NFL Big Data Bowl 2023*. Indianapolis, IN. March 2023. (Invited)

**Finalist (Top 8 + \$10,000 prize + Presented at 2023 NFL Scouting Combine)**

**Q. Nguyen**, G. J. Matthews. Estimating Aging Curves: Using Multiple Imputation to Examine Career Trajectories of MLB Offensive Players. *Carnegie Mellon Sports Analytics Conference*. Pittsburgh, PA. October 2022. (Invited)

**Finalist, Open Methodology Track, Reproducible Research Competition**

**Q. Nguyen**, H. Butler, G. J. Matthews. An Examination of Olympic Sport Climbing Competition Format and Scoring System. *Cascadia Symposium on Statistics in Sports*. Vancouver, BC. September 2022. (Invited)

**Q. Nguyen**, H. Butler, G. J. Matthews. An Examination of Sport Climbing's Competition Format and Scoring System. *Carnegie Mellon Sports Analytics Conference*. Pittsburgh, PA. November 2021. (Invited, virtual talk)

**Winner, Open Methodology Track, Reproducible Research Competition**

### POSTERS

**Q. Nguyen**, H. Butler, G. J. Matthews. Concerns Regarding Sport Climbing Competition Format and Scoring System: Who's Suffering? *UConn Sports Analytics Symposium*. October 2021. (Virtual)

**Winner, Student Poster Award**

**Q. Nguyen.** Poisson Modeling and Predicting English Premier League Goal Scoring. *NextGen: Data Science Day, New England Statistical Society*. November 2020. (Virtual)

**Co-Winner, Poster Competition, Undergrad/MS Group**

**Q. Nguyen.** Poisson Modeling and Predicting English Premier League Goal Scoring. *Carnegie Mellon Sports Analytics Conference*. October 2020. (Virtual)

**Q. Nguyen.** Poisson Modeling and Predicting English Premier League Goal Scoring. *UConn Sports Analytics Symposium*. October 2020. (Virtual)

## TUTORIALS & WORKSHOPS

**Q. Nguyen.** Unsupervised learning with tracking data (Advanced level). *Connecticut Sports Analytics Symposium*. Storrs, CT. April 2026.

**Q. Nguyen.** Rcpp: A primer. *StatBytes Statistical Computing Seminars (Department of Statistics & Data Science, Carnegie Mellon University)*. Pittsburgh, PA. November 2025.

**Q. Nguyen.** Building a performance metric with player tracking data (Advanced level). *Connecticut Sports Analytics Symposium*. New Haven, CT. April 2025.

**Q. Nguyen.** Get Trained & Get Strained: The Ingredients of a Big Data Bowl Metric. *Carnegie Mellon Sports Analytics Conference*. Pittsburgh, PA. November 2023.

## GUEST LECTURES

**Q. Nguyen.** Bayesian multilevel modeling in the wild: Beyond homogeneity. *Special Topics: Sports Analytics (36-460/660, Carnegie Mellon University)*. Pittsburgh, PA. March 2025 & March 2026.

**Q. Nguyen.** Big Data Bowl: Overview, Getting Started, My Experience, and more. *Summer Undergraduate Research Experience in Statistics (Carnegie Mellon Sports Analytics Camp)*. Pittsburgh, PA. July 2023.

**Q. Nguyen.** Web Scraping: A Primer. *Summer Undergraduate Research Experience in Statistics (Carnegie Mellon Sports Analytics Camp)*. Pittsburgh, PA. July 2023.

## HONORS & AWARDS

|                                                                                                                             |                  |
|-----------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Student Travel Support</b> , Connecticut Sports Analytics Symposium                                                      | 2024, 2025, 2026 |
| <b>Sports Analytics Student Award</b> , Carnegie Mellon Statistics & Data Science<br>Funded by the Richardson Matano Family | 2025             |
| <b>NFL Big Data Bowl Semifinalist</b>                                                                                       | 2025             |
| <b>NFL Big Data Bowl Finalist</b>                                                                                           | 2023, 2024       |
| <b>ASA Section on Statistics in Sports Student Funding</b>                                                                  | 2024             |
| <b>Graduate Student Assembly/Provost Conference Funds</b> , Carnegie Mellon University                                      | 2024             |
| <b>Reproducible Research Competition Finalist</b> , Carnegie Mellon Sports Analytics Conference                             | 2022             |
| <b>University of Toronto Sports Analytics Group Scholarship</b> , Cascadia Symposium on Statistics in Sports                | 2022             |
| <b>Reproducible Research Competition Winner</b> , Carnegie Mellon Sports Analytics Conference                               | 2021             |
| <b>Student Poster Award Winner</b> , UConn Sports Analytics Symposium                                                       | 2021             |
| <b>Merit Award</b> , MS in Applied Statistics Program, Loyola University Chicago                                            | 2020 – 2021      |
| <b>Poster Competition Co-Winner</b> , NextGen: Data Science Day, New England Statistical Society                            | 2020             |
| <b>Summa Cum Laude</b> , Wittenberg University                                                                              | 2020             |
| <b>Departmental Honors</b> , Department of Mathematics and Computer Science, Wittenberg University                          | 2020             |

|                                                                                                                                                                                                                   |                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Phi Beta Kappa</b> , Omicron of Ohio Chapter, Wittenberg University                                                                                                                                            | 2020                       |
| <b>Pi Mu Epsilon</b> , Ohio Upsilon Chapter, Wittenberg University                                                                                                                                                | 2020                       |
| <b>The Charles and Elsie Little Mathematics Award</b><br>Department of Mathematics and Computer Science, Wittenberg University<br>Awarded to the most outstanding senior majoring in mathematics and data science | 2020                       |
| <b>Mortar Board</b> , Wittenberg University                                                                                                                                                                       | 2020                       |
| <b>Pick and Pen</b> , Wittenberg University                                                                                                                                                                       | 2019                       |
| <b>Mathematics 202 Award</b> , Department of Mathematics and Computer Science, Wittenberg University                                                                                                              | 2018                       |
| <b>Textbook Award</b> , Department of Mathematics and Computer Science, Wittenberg University                                                                                                                     | 2018                       |
| <b>Phi Eta Sigma</b> , Wittenberg University                                                                                                                                                                      | 2018                       |
| <b>Dean's List</b> , Wittenberg University                                                                                                                                                                        | All semesters; 2017 – 2020 |

## SERVICES

|                                                                                                                                                                                                         |                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Organizer</b><br>Carnegie Mellon Sports Analytics Center (CMSAC) Research Lab                                                                                                                        | 2023 – present                         |
| <b>Program Committee</b><br>UConn/Connecticut Sports Analytics Symposium                                                                                                                                | 2024, 2025, 2026                       |
| <b>Reviewer</b><br>Journal of Quantitative Analysis in Sports, Machine Learning, The American Statistician, WIREs Computational Statistics                                                              |                                        |
| <b>Judge</b><br>CMSAC Reproducible Research Competition<br>CSAS Student Poster Award<br>CSAS Data Challenge                                                                                             | 2024, 2025<br>2024, 2025<br>2024, 2025 |
| <b>Mentoring</b><br>Women in Data Science (WiDS) Chicago Datathon                                                                                                                                       | 2022                                   |
| <b>Statistical Consulting</b><br>Carnegie Mellon University Athletics<br>Loyola University Chicago<br>(Clients: Stritch School of Medicine; Department of Biology; City of Milwaukee Assessor's Office) | 2024<br>2021 – 2022                    |

## MEDIA

|                                                                                                                                |  |
|--------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Coverage &amp; Mentions</b><br>NBC Sports, Carnegie Mellon University, Knowable Magazine, Amstat News, ABC News (Australia) |  |
| <b>Podcast &amp; Show Appearances</b><br>Open Source Sports Podcast, Viz Buzz (data science game show; multiple seasons)       |  |

## SKILLS

Computing: R/RStudio (expert); Python, SQL, SAS, Git/GitHub (intermediate)  
Typesetting: markdown,  $\text{\LaTeX}$  (proficient)  
Bilingual: Vietnamese (native language) and English (proficient)

*Last updated June 8, 2026*